

Enhancing Health Management Through Wearable Devices

2.1 Problem Statement

Team Name: RCJ

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Introduction and Motivation:

Wearable devices like Apple Watch, Fitbit, and Garmin offer rich health and fitness tracking, but many users have a hard time understanding the data, so they don't use these features fully.

Problem Statement:

Despite purchasing smartwatches for health monitoring, users often miss key insights due to difficulties engaging with and understanding the provided data, which limits proactive health management.

HCI Challenge:

The key challenge is to create an intuitive interface that transforms complex wearable data into clear, actionable insights. This would ultimately boost user engagement and promote healthier lifestyles.

2.2 Research Question and Hypothesis

Research Question:

How can wearable health device interfaces be improved to enhance user engagement and promote better health outcomes?

Hypothesis:

More personalized and user-friendly health-tracking interfaces will increase user engagement and improve healthy behaviors.

Literature Review:

We will review 10-15 relevant studies on wearable health technology, user engagement, and HCI best practices.

2.3 Target Users and Use Cases (Assumptions)

Primary Users:

- Individuals actively trying to improve their fitness and health.
- Patients with chronic conditions who need continuous monitoring.

Secondary Users:

- Healthcare providers seeking remote monitoring solutions.
- Researchers studying digital health interventions.

Use Cases:

- A user monitors their heart rate and receives recommendations to manage stress.
- A diabetic patient uses blood glucose tracking to adjust their diet.

2.4 Initial Market Research

Competitor Analysis:

- Apple Watch: Comprehensive health features but can be overwhelming for users.
- Fitbit: User-friendly but lacks deeper health insights.
- Garmin: These are advanced metrics for athletes but may be too technical for casual users.

User Pain Points:

- Confusing data presentation.
- Inconsistent engagement with health apps.

Industry Trends:

- Increased integration of AI-driven health recommendations.
- Growing emphasis on mental health tracking in wearables.

2.5 Proposed Methodology (for doing User Research)

User Research Methods:

- Surveys to assess user engagement with wearable health devices.
- Usability testing with prototypes featuring improved data visualization.

Justification:

These methods will provide a comprehensive understanding of user needs and preferences, guiding interface improvements.

2.6 Expected Outcomes

Research Outcomes:

- Insights into user behavior and engagement barriers.
- Recommendations for improving wearable device interfaces.

Product Outcomes:

- A prototype of an improved health-tracking interface.
- Features include personalized notifications, intuitive data visualization, and actionable health recommendations.

Impact:

- Better health management and lifestyle improvements.
- Contribution to HCI and digital health research.

2.7 AI Disclosure & Contribution Statement

AI Disclosure:

Research question from ChatGPT.

Layout and simplify using ChatGPT and Grammarly.

Contribution Statement:

Hsueh-Ta Lu: led market research and competitor analysis, identifying user pain points and industry trends, and refining the research question, hypothesis, and expected outcomes.

Tung-Ying Lee: focused on user research, designing surveys and usability tests to assess engagement, analyzed user data, and translated findings into recommendations.

Po-Han Chen: managed the HCI framework and usability improvements, ensuring alignment with best practices.

Hsin-Ying Tsai: handled interface design and prototyping, creating intuitive data visualizations and personalized notifications.